QUANT DEV / C++ / SYSTEMS

EXCHANGE LAB

PROJECT 06

Low latency systems engineering with correctness first.

A deterministic C++ limit order book matching engine used to examine the tradeoff between price time correctness, memory layout and tail latency.

PROPOSED RESEARCH / NOT YET VERIFIED

SYSTEM / RESEARCH ARCHITECTURE

1 STAGE
Order generator

→

2 STAGE
Risk + sequence

→

3 STAGE
Matching engine

→

4 STAGE
Journal + replay

THE PROBLEM TO SOLVE

What optimisations improve event processing latency without breaking matching determinism or accounting invariants?

TECHNICAL DISTINCTION

1. A basic order book is not enough; the value lies in rigorous testing and defensible benchmarking.
2. You need a simple reference model, adversarial order flow and careful measurement discipline.

WHAT THE EVIDENCE MUST SHOW

- ✓ Matching invariants across adversarial sequences
- ✓ Documented p50 / p99 / p99.9 latency under fixed workloads
- ✓ Memory use and like for like baseline comparison